

B.Tech. Degree III Semester Regular/Supplementary Examination November 2023

IT 19-204-0302 DIGITAL ELECTRONICS (2019 Scheme)

Time: 3 Hours

Maximum Marks: 60

Course Outcome

On successful completion of the course, the students will be able to:

CO1: Represent different number systems, circuits, logic gates

CO2: Design Combinational and Sequential circuits.

CO3: Analyse Logic families

CO4: Familiarized with the basic principles of memory.

Bloom's Taxonomy Levels (BL): L1 – Remember, L2 – Understand, L3 – Apply, L4 – Analyze, L5 – Evaluate, L6 – Create

PO – Programme Outcome

PART A

(Answer ALL questions)

	(8 × 3 = 24)	Marks	BL	CO	PO
I. (a) State:		3	L1	1	1
(i) Consensus Law					
(ii) De-Morgan Theorem					
(iii) Principle of Duality.					
(b) (i) Perform $(185)_{10} - (8)_{10}$ using excess-3 code.		3	L3	1	2
(ii) Perform addition of decimal numbers +38 and -18 using the 2's complement method.					
(c) Design 4-bit Ripple carry adder.		3	L2	2	3
(d) Explain the design procedure of combinational circuit. Give example.		3	L2	2	1
(e) Differentiate between combinational circuit and sequential circuit with example.		3	L4	2	1
(f) Design asynchronous mod-10 counter.		3	L6	2	4
(g) Explain triggering methods of Flipflops.		3	L4	2	2
(h) Define:		3	L2	3	1
(i) Propagation delay					
(ii) Noise Margins					
(iii) Fan in and Fan out.					

PART B

(4 × 12 = 48)

II. (a)	$F(A, B, C) = \sum m(2, 3, 4, 5)$ and $\sum d(6, 7)$ simplify using K-Map and implement the minimal expression in universal logic.	6	L3	1	2
(b)	Perform the conversions:	6	L3	1	2
(i)	$(259A)_{16} = ()_2$				
(ii)	$(37.45)_8 = ()_2$				
(iii)	$(CAFÉ.3D)_{16} = ()_2$				
(iv)	$(10110.11)_2 = ()_8$				
(v)	$(CAD)_{16} = ()_8$				
(vi)	$(25.625)_{10} = ()_8$				

OR

(P.T.O.)

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		Marks	BL	CO	PO
III.	(a) $F = \sum m(0,1,3,7,8,9,11,15)$ solve using Quine McClusky technique.	6	L3	1	1
	(b) Minimize the Boolean expression for				
	(i) $F(x, y) = \sum m(0,2,3,6,7)$	3	L3	1	1
	(ii) $F(x, y, z) = \sum m(0,1,2,5,7,8,9,10,13,15)$.	3	L3	1	1
IV.	(a) Design:	6	L6	2	3
	(i) 8:1 Mux				
	(ii) 1:4 Demux.				
	(b) Implement Boolean function using multiplexer	6	L3	1	1
	$F(A, B, C, D) = \sum (0,1,2,3,4,9,13,14,15)$				
	(i) 8:1 mux				
	(ii) 16:1 mux.				
OR					
V.	(a) Draw the block diagram for the following:	2	L3	2	2
	(i) 4-bit Ripple Carry Adder	2	L3	2	2
	(ii) 3-bit subtractor	2	L3	2	2
	(iii) 5-bit Adder/subtractor Circuit.				
	(b) Design:				
	(i) 4:2 Encoder with enable input	2	L6	2	3
	(ii) 4:2 Priority Encoder (Inputs : I0,I1,I2,I3 ; and with priority order consider I3 as highest priority and I0 has lowest priority)	2	L6	2	3
	(iii) 3: 8 Decoder with no enable input.	2	L6	2	3
VI.	Design:				
	(i) 3-bit synchronous up counter	4	L6	2	3
	(ii) 3-bit synchronous down counter	4	L6	2	3
	(iii) 3-bit asynchronous updown counter.	4	L6	2	3
OR					
VII.	Explain Shift Register and types of registers in detail:	12	L2	2	5
	(i) SISO				
	(ii) PISO				
	(iii) PIPO				
	(iv) SIPO				
VIII.	(a) Draw and explain the structure of PLA.	6	L2	4	1
	(b) Differentiate between ROM and RAM.	6	L2	4	1
OR					
IX.	Explain:				
	(i) MOS and CMOS	6	L2	3	1
	(ii) Differentiate between RTL and DTL.	6	L2	3	1

Blooms's Taxonomy Levels

L1 – 2.5%, L2 – 37.5%, L3 – 32.5%, L4 – 5%, L6 – 22.5%.
